NT-4b: Literature Extraction & Convention Fixing for Full Nonlinear Variational Field Equations

From Nonlocal Effective Action to $\frac{\delta S}{\delta g^{\mu\nu}} = 0$ on Curved Backgrounds

David Alfyorov

March 2026 — Working Document

Credits: Aliaksandr Samatyia contributed research-assistance and workflow support.

Abstract

This document collects all formulas, conventions, and computational recipes from the literature needed to derive the full nonlinear gravitational field equations from the SCT spectral action $\mathrm{Tr}(f(D^2/\Lambda^2))$ on arbitrary curved backgrounds. Unlike the NT-4a linearization (flat background, linear perturbation), NT-4b requires the variation of a nonlocal curvature-squared action $\delta S/\delta g^{\mu\nu}$ where the form factors $F_i(\Box/\Lambda^2)$ do not commute with the metric variation. The document covers: (1) the SCT effective action and its conventions inherited from NT-1 through NT-4a; (2) the local Stelle field equations as the limiting benchmark; (3) the Barvinsky–Vilkovisky formalism for varying nonlocal actions; (4) the nonlocal Bach tensor and nonlocal R^2 variation; (5) key identities for $\delta\Box/\delta g^{\mu\nu}$ and the α-insertion formula; (6) consistency checks (Bianchi identity, trace, linearized limit, de Sitter solutions); (7) a comprehensive notation table. All formulas are extracted from the literature with source annotations; no original derivations are performed. Gaps requiring resolution in the derivation phase are noted explicitly.

Key references:

- Stelle: Phys. Rev. D **16** (1977) 953; Gen. Rel. Grav. **9** (1978) 353.
- Barvinsky–Vilkovisky (BV): Phys. Rep. **119** (1985) 1; Nucl. Phys. B **282** (1987) 163; B **333** (1990) 471.
- Codello–Zanusso (CZ): arXiv:1203.2034 [hep-th].
- Biswas–Mazumdar–Siegel (BMS): hep-th/0508194.
- Modesto: arXiv:1107.2403 [hep-th].
- Calcagni–Modesto–Nardelli (CMN): arXiv:1803.07848; also Calcagni–Modesto–Leszczyszyn: arXiv:1803.00561.
- Buoninfante–Modesto–Shapiro: arXiv:2001.07830.
- Giacchini–de Paula Netto: arXiv:1806.05664 [gr-qc]; Eur. Phys. J. C **81** (2021) 741.
- Deser–Woodard: Phys. Rev. Lett. **99** (2007) 111301.
- Avramidi: "Heat Kernel and Quantum Gravity," Springer (2000).

- Parker–Toms: "Quantum Field Theory in Curved Spacetime," Cambridge (2009).

Contents

1 Background: From NT-4a to NT-4b **4**
- 1.1 What NT-4a established . . . 4
- 1.2 What NT-4b must achieve . . . 4
- 1.3 Gap analysis from NT-4a . . . 4

2 The SCT Effective Action: Conventions **4**
- 2.1 Sign and curvature conventions . . . 4
- 2.2 The full SCT action . . . 5
- 2.3 Input coefficients from Phase 3 . . . 6
- 2.4 The normalized form factors . . . 6

3 Local Benchmark: Stelle's Nonlinear Field Equations **6**
- 3.1 The Stelle action and field equations . . . 7
- 3.2 Variation of the Gauss–Bonnet term . . . 7
- 3.3 Individual curvature-squared variations (local) . . . 8

4 Nonlocal Variation: The Central Problem **8**
- 4.1 Why the variation is nontrivial . . . 8
- 4.2 Variation of $\sqrt{g}$ . . . 9
- 4.3 Variation of the covariant d'Alembertian . . . 9

5 The α-Insertion Formula (Barvinsky–Vilkovisky) **9**
- 5.1 The fundamental identity . . . 9
- 5.2 Variation of $R\,F(\Box)\,R$ . . . 10
- 5.3 Variation of $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}F(\Box)\,C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ (nonlocal Bach) . . . 11

6 Nonlocal Gravity in the $G_{\mu\nu}\gamma(\Box)R^{\mu\nu}$ Basis **11**
- 6.1 The simplified action . . . 11
- 6.2 Mapping to the SCT $\{C^2, R^2\}$ basis . . . 12

7 Key Identities for the Nonlinear Variation **13**
- 7.1 Commutator of $\Box$ with curvature tensors . . . 13
- 7.2 The Lichnerowicz operator . . . 13
- 7.3 Integration by parts with $F(\Box)$ . . . 13

8 Consistency Checks for the Full Nonlinear Equations **14**
- 8.1 Bianchi identity . . . 14
- 8.2 Trace equation . . . 14
- 8.3 Linearized limit . . . 14
- 8.4 De Sitter solutions . . . 15

9 Auxiliary Field Formulation **15**

10 Notation Table **16**

11 Gap Analysis **17**

12 Strategy for the NT-4b Derivation **17**

1 Background: From NT-4a to NT-4b

1.1 What NT-4a established

NT-4a derived the *linearized* field equations of the SCT spectral action on a flat background. The key results are:

- **Modified propagator:** $G_{\mathrm{TT}}^{-1} = k^2\,\Pi_{\mathrm{TT}}(z)$ with $\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\,z\,\hat{F}_1(z)$.
- **Scalar mode:** $\Pi_s(z,\xi) = 1 + 6(\xi - 1/6)^2\,z\,\hat{F}_2(z,\xi)$.
- **Yukawa potential:** $V(r)/V_N(r) = 1 - \frac{4}{3}e^{-m_2 r} + \frac{1}{3}e^{-m_0 r}$.
- **Gauge invariance** and **Bianchi identity** verified off-shell.
- **40/40 checks PASS**, 7 Lean 4 identities verified.

All these results are *valid only on flat background at linear order*.

1.2 What NT-4b must achieve

NT-4b must derive the **full nonlinear** field equations

$$\frac{1}{\sqrt{g}}\,\frac{\delta S_{\mathrm{SCT}}}{\delta g^{\mu\nu}} = 0 \tag{1}$$

on *arbitrary* curved backgrounds, where S_{SCT} includes both the Einstein–Hilbert term and the nonlocal curvature-squared corrections.

The central difficulty is that the form factors $F_i(\Box/\Lambda^2)$ are functions of the covariant d'Alembertian, which itself depends on the metric. Therefore $[\delta/\delta g^{\mu\nu},\, F(\Box)] \neq 0$, and the variation generates additional "Θ-type" terms absent in the local theory.

1.3 Gap analysis from NT-4a

The NT-4a literature document [23] identified the following gaps relevant to NT-4b:

G1. Full nonlinear equations on curved background (Barvinsky–Vilkovisky formalism) — **this document**.

G2. FLRW reduction — deferred to NT-4c.

G3. Covariant gauge-fixing analysis — needed for NT-4b but secondary.

2 The SCT Effective Action: Conventions

2.1 Sign and curvature conventions

Convention 2.1 (Curvature conventions)**.** [our convention] Throughout this project, the conventions are:

Quantity	Convention
Metric signature	$(-,+,+,+)$ (Lorentzian) or Euclidean $\delta_{\mu\nu}$
Riemann tensor	$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$
Ricci tensor	$R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}$
Ricci scalar	$R = g^{\mu\nu} R_{\mu\nu}$
Einstein tensor	$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$
d'Alembertian	$\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu = \frac{1}{\sqrt{\lvert g\rvert}}\partial_\mu(\sqrt{\lvert g\rvert}\, g^{\mu\nu}\partial_\nu)$ on scalars
Weyl tensor	$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} + \frac{2}{d-2}(g_{\nu[\rho}R_{\sigma]\mu} - g_{\mu[\rho}R_{\sigma]\nu}) + \frac{2}{(d-1)(d-2)} R\, g_{\mu[\rho}g_{\sigma]\nu}$
Gravitational coupling	$\kappa^2 = 8\pi G$

These match the conventions used in Calcagni–Modesto [11], Stelle [1, 2], and all preceding NT-1/NT-1b/NT-2/NT-4a documents.

2.2 The full SCT action

Convention 2.2 (Full SCT gravitational action)**.** [our convention] The full one-loop SCT gravitational action is:

$$S = S_{\text{EH}} + S_{\text{quad}} + S_{\text{matter}}, \tag{2}$$

where:

$$S_{\text{EH}} = \frac{1}{2\kappa^2}\int \mathrm{d}^4x\,\sqrt{\lvert g\rvert}\,(R - 2\Lambda_{\text{cc}}), \tag{3}$$

$$S_{\text{quad}} = \frac{1}{16\pi^2}\int \mathrm{d}^4x\,\sqrt{\lvert g\rvert}\left[F_1\left(\frac{\Box}{\Lambda^2}\right) C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + F_2\left(\frac{\Box}{\Lambda^2},\xi\right) R^2\right], \tag{4}$$

$$S_{\text{matter}} = \int \mathrm{d}^4x\,\sqrt{\lvert g\rvert}\,\mathcal{L}_{\text{matter}}(g_{\mu\nu},\psi). \tag{5}$$

The cosmological constant term Λ_{cc} comes from the a_0 Seeley–DeWitt coefficient; we keep it general here.

Remark 2.3 (Curvature basis choice). The $\{C^2, R^2\}$ basis is preferred over $\{R^2_{\mu\nu}, R^2\}$ because:

1. $C_{\mu\nu\rho\sigma}$ is traceless, so C^2 projects onto pure spin-2.
2. The Gauss–Bonnet identity in $d = 4$:

$$E_4 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2 \tag{6}$$

   is topological ($\int \sqrt{g}\, E_4 = 32\pi^2\chi(M)$) and does not contribute to the equations of motion.
3. Conversion:

$$C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 2R_{\mu\nu}R^{\mu\nu} + \tfrac{1}{3}R^2. \tag{7}$$

Remark 2.4 (Alternative Ricci–scalar basis). [Calcagni–Modesto 2211.05606, Eq. (NLGActionGeneral)] The general nonlocal gravity action in the $\{R^2_{\mu\nu\rho\sigma}, R^2_{\mu\nu}, R^2\}$ basis reads:

$$S_{\text{NLG}} = \frac{1}{2\kappa^2}\int \mathrm{d}^4x\,\sqrt{\lvert g\rvert}\left[R + R\,\gamma_0(\Box)\,R + R_{\mu\nu}\,\gamma_2(\Box)\,R^{\mu\nu} + R_{\mu\nu\rho\sigma}\,\gamma_4(\Box)\,R^{\mu\nu\rho\sigma}\right]. \tag{8}$$

Using the Gauss–Bonnet identity (6) to eliminate $R^2_{\mu\nu\rho\sigma}$, this becomes:

$$S = \frac{1}{2\kappa^2}\int \mathrm{d}^4x\,\sqrt{|g|}\left[R + R\,\tilde{\gamma}_0(\Box)\,R + R_{\mu\nu}\,\tilde{\gamma}_2(\Box)\,R^{\mu\nu}\right] + \text{topological}, \tag{9}$$

with $\tilde{\gamma}_0 = \gamma_0 - \gamma_4$ and $\tilde{\gamma}_2 = \gamma_2 + 4\gamma_4$.

To map to our $\{C^2, R^2\}$ basis, one uses (7) to express:

$$\alpha\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \beta\, R^2 = \alpha\, R^2_{\mu\nu\rho\sigma} - 2\alpha\, R_{\mu\nu}R^{\mu\nu} + \left(\beta + \tfrac{\alpha}{3}\right) R^2. \tag{10}$$

Matching: $\gamma_4 = \alpha$, $\gamma_2 = -2\alpha$, $\gamma_0 = \beta + \alpha/3$.

2.3 Input coefficients from Phase 3

[NT-1b Phase 3]

$$\alpha_C = 16\pi^2 F_1(0) = \frac{13}{120}, \tag{11}$$

$$\alpha_R(\xi) = 16\pi^2 F_2(0,\xi) = 2\left(\xi - \frac{1}{6}\right)^2. \tag{12}$$

In the $\{R^2, R^2_{\mu\nu}\}$ basis:

$$c_2 = 2\alpha_C = \frac{13}{60}, \qquad 3c_1 + c_2 = 6\left(\xi - \frac{1}{6}\right)^2. \tag{13}$$

[Verified] $\alpha_C = 13/120$ and $\alpha_R(\xi) = 2(\xi - 1/6)^2$ confirmed by NT-1b Phase 3 (354/354 checks PASS).

2.4 The normalized form factors

Definition 2.5 (Normalized form factors)**.** [our convention]

$$\hat{F}_i(z) = \frac{F_i(z)}{F_i(0)}, \qquad \hat{F}_i(0) = 1. \tag{14}$$

The master function underlying all form factors is:

$$\varphi(z) = e^{-z/4}\sqrt{\frac{\pi}{z}}\,\mathrm{erfi}\left(\frac{\sqrt{z}}{2}\right), \qquad \varphi(0) = 1, \quad \varphi'(0) = -\frac{1}{6}. \tag{15}$$

The Taylor coefficients are $a_n = (-1)^n n!/(2n+1)!$ (NT-2, verified entire).

3 Local Benchmark: Stelle's Nonlinear Field Equations

The local limit $F_1 \to F_1(0)$, $F_2 \to F_2(0)$ must reproduce Stelle's fourth-order gravity. This section documents the exact nonlinear field equations from Stelle's 1977–1978 papers [1, 2].

3.1 The Stelle action and field equations

Convention 3.1 (Stelle action). [Stelle, Phys. Rev. D **16** (1977) 953; Gen. Rel. Grav. **9** (1978) 353]

$$S_{\text{Stelle}} = \frac{1}{2\kappa^2} \int \mathrm{d}^4x \sqrt{|g|} \left[R + \alpha\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} + \beta\, R^2 \right]. \tag{16}$$

Theorem 3.2 (Stelle's nonlinear field equations). *[Stelle 1978, Eqs. (3.2)–(3.5)] The metric variation $\delta S_{\text{Stelle}}/\delta g^{\mu\nu} = 0$ gives:*

$$\boxed{G_{\mu\nu} + 2\alpha\, B_{\mu\nu} + \beta\, H_{\mu\nu} = \kappa^2 T_{\mu\nu},} \tag{17}$$

*where the **Bach tensor** $B_{\mu\nu}$ and the **scalar curvature tensor** $H_{\mu\nu}$ are:*

$$B_{\mu\nu} = \nabla^\rho \nabla^\sigma C_{\mu\rho\nu\sigma} + \frac{1}{2} R^{\rho\sigma} C_{\mu\rho\nu\sigma}, \tag{18}$$

$$H_{\mu\nu} = 2\nabla_\mu \nabla_\nu R - 2 g_{\mu\nu} \Box R - \frac{1}{2} g_{\mu\nu} R^2 + 2R\, R_{\mu\nu}. \tag{19}$$

Remark 3.3 (Alternative forms of the Bach tensor). [Stelle 1978; Bach 1921] Using the Bianchi identity, the Bach tensor can be rewritten as:

$$B_{\mu\nu} = \Box R_{\mu\nu} - \frac{1}{3} \nabla_\mu \nabla_\nu R - \frac{1}{6} g_{\mu\nu} \Box R + R_{\mu\rho\nu\sigma} R^{\rho\sigma} - \frac{1}{3} R\, R_{\mu\nu} - \frac{1}{4} g_{\mu\nu} \left(R_{\mu\nu} R^{\mu\nu} - \frac{1}{3} R^2 \right). \tag{20}$$

This form makes the fourth-derivative (quasi-linear) structure manifest: $B_{\mu\nu}$ contains $\Box R_{\mu\nu}$ (four derivatives of g) as its leading term.

The Bach tensor is traceless in $d = 4$:

$$g^{\mu\nu} B_{\mu\nu} = 0. \tag{21}$$

This follows from the conformal invariance of $\int \sqrt{g}\, C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}$.

Remark 3.4 (Properties of $H_{\mu\nu}$). [Stelle 1978] The trace of $H_{\mu\nu}$ is:

$$g^{\mu\nu} H_{\mu\nu} = -6 \Box R. \tag{22}$$

The divergence satisfies:

$$\nabla^\mu H_{\mu\nu} = 0, \tag{23}$$

which ensures $\nabla^\mu (G_{\mu\nu} + 2\alpha B_{\mu\nu} + \beta H_{\mu\nu}) = 0$ (Bianchi identity for the full equations).

[Sign correction] Some references define $B_{\mu\nu}$ with the opposite sign convention. Our convention follows Stelle (1978) and Parker–Toms (2009), where $\delta(C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma})/\delta g^{\mu\nu} = 2B^{\mu\nu}$ (with the factor 2 absorbed into the coefficient α). Other references (e.g., Oliva–Ray 2010) define $B_{\mu\nu}$ such that $\delta(C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma})/\delta g^{\mu\nu} = -2B^{\mu\nu}$. The sign must be tracked carefully in NT-4b.

3.2 Variation of the Gauss–Bonnet term

Proposition 3.5 (Gauss–Bonnet variation in $d = 4$). *[exact from lit.] [Padmanabhan, "Gravitation," Ch. 15; Lovelock 1971] In $d = 4$, the Gauss–Bonnet combination E_4 is topological:*

$$\frac{\delta}{\delta g^{\mu\nu}} \int \mathrm{d}^4x \sqrt{|g|}\, E_4 = 0. \tag{24}$$

This identity is crucial: it means that any basis conversion between $\{R^2_{\mu\nu\rho\sigma}, R_{\mu\nu} R^{\mu\nu}, R^2\}$ and $\{C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma}, R^2\}$ does not generate additional equations-of-motion contributions.

3.3 Individual curvature-squared variations (local)

For reference and as building blocks for the nonlocal case, we record the local metric variations of each curvature-squared invariant.

Proposition 3.6 (Metric variation of curvature-squared terms)**.** *[Padmanabhan Ch. 15; Parker–Toms App. B; Boulware–Deser, Phys. Rev. Lett.* ***55*** *(1985) 2656]*

$$\frac{1}{\sqrt{g}}\frac{\delta(\sqrt{g}\,R^2)}{\delta g^{\mu\nu}} = 2\nabla_\mu\nabla_\nu R - 2g_{\mu\nu}\Box R + 2R\,R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R^2 \equiv H_{\mu\nu}, \tag{25}$$

$$\frac{1}{\sqrt{g}}\frac{\delta(\sqrt{g}\,R_{\mu\nu}R^{\mu\nu})}{\delta g^{\mu\nu}} = \Box R_{\mu\nu} + \frac{1}{2}g_{\mu\nu}\Box R - \nabla_\mu\nabla_\nu R + 2R_{\mu\rho\nu\sigma}R^{\rho\sigma} - \frac{1}{2}g_{\mu\nu}R_{\mu\nu}R^{\mu\nu}, \tag{26}$$

$$\frac{1}{\sqrt{g}}\frac{\delta(\sqrt{g}\,R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})}{\delta g^{\mu\nu}} = 4\Box R_{\mu\nu} + 4g_{\mu\nu}\Box R - 6\nabla_\mu\nabla_\nu R + 8R_{\mu\rho\nu\sigma}R^{\rho\sigma} - 2g_{\mu\nu}R_{\mu\nu}R^{\mu\nu} + \frac{1}{2}g_{\mu\nu}R^2 - 2R\,R_{\mu\nu}. \tag{27}$$

Remark 3.7 (Relation between variations). Using the Gauss–Bonnet identity (24), the three variations (25)–(27) satisfy:

$$\frac{\delta(\sqrt{g}\,R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})}{\delta g^{\mu\nu}} - 4\frac{\delta(\sqrt{g}\,R_{\mu\nu}R^{\mu\nu})}{\delta g^{\mu\nu}} + \frac{\delta(\sqrt{g}\,R^2)}{\delta g^{\mu\nu}} = 0. \tag{28}$$

This provides a consistency check on all three formulas.

4 Nonlocal Variation: The Central Problem

4.1 Why the variation is nontrivial

For a local action $S = \int \sqrt{g}\,A\cdot B$ where A and B are curvature scalars, the variation is simply $\delta S/\delta g^{\mu\nu} = (\delta A/\delta g^{\mu\nu})\cdot B + A\cdot(\delta B/\delta g^{\mu\nu}) + \dots$

For a *nonlocal* action

$$S = \int \sqrt{g}\,A\,F(\Box)\,B, \tag{29}$$

the variation generates an additional term because $\Box$ depends on $g^{\mu\nu}$:

$$\frac{\delta}{\delta g^{\mu\nu}}\Big[A\,F(\Box)\,B\Big] = \frac{\delta A}{\delta g^{\mu\nu}}\,F(\Box)\,B + A\,F(\Box)\,\frac{\delta B}{\delta g^{\mu\nu}} + A\,\frac{\delta F(\Box)}{\delta g^{\mu\nu}}\,B. \tag{30}$$

The third term requires computing $\delta F(\Box)/\delta g^{\mu\nu}$, which involves $\delta\Box/\delta g^{\mu\nu}$. Since F is a function (not just $\Box$ itself), this requires the "α-insertion formula" from the Barvinsky–Vilkovisky formalism.

4.2 Variation of $\sqrt{g}$

Proposition 4.1 (Fundamental metric variations). *[exact from lit.]*

$$\frac{\delta g_{\alpha\beta}}{\delta g^{\mu\nu}} = -\frac{1}{2}(g_{\alpha\mu}g_{\beta\nu} + g_{\alpha\nu}g_{\beta\mu}), \tag{31}$$

$$\frac{\delta\sqrt{|g|}}{\delta g^{\mu\nu}} = -\frac{1}{2}\sqrt{|g|}\, g_{\mu\nu}, \tag{32}$$

$$\frac{\delta R}{\delta g^{\mu\nu}} = R_{\mu\nu} + g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu, \tag{33}$$

$$\begin{aligned}\frac{\delta R_{\alpha\beta}}{\delta g^{\mu\nu}} = \frac{1}{2}\Big(&g_{\alpha\mu}\,\nabla_\nu\nabla_\beta + g_{\alpha\nu}\,\nabla_\mu\nabla_\beta + \nabla_\alpha\nabla_\mu\, g_{\beta\nu} + \nabla_\alpha\nabla_\nu\, g_{\beta\mu}\\ &- \Box\, g_{\alpha(\mu}g_{\nu)\beta} - g_{\alpha\beta}\nabla_\mu\nabla_\nu - R_{\alpha(\mu\nu)\beta}\Big).\end{aligned} \tag{34}$$

These are standard identities; see Parker–Toms [19] App. B.

4.3 Variation of the covariant d'Alembertian

Proposition 4.2 (Variation of $\Box$ acting on a scalar ϕ). *[Barvinsky–Vilkovisky, Phys. Rep.* ***119*** *(1985); Deser–Woodard 2007]*

$$\delta[\Box\phi] = (\delta g^{\mu\nu})\nabla_\mu\nabla_\nu\phi + g^{\mu\nu}\,\delta\Gamma^\rho_{\mu\nu}\,\nabla_\rho\phi, \tag{35}$$

where

$$\delta\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}\Big(\nabla_\mu\delta g_{\nu\sigma} + \nabla_\nu\delta g_{\mu\sigma} - \nabla_\sigma\delta g_{\mu\nu}\Big). \tag{36}$$

For the specific variation $\delta g^{\mu\nu} = h^{\mu\nu}$ and working to all orders on curved background, the key identity is:

$$\frac{\delta(\Box\phi)}{\delta g^{\mu\nu}} = \nabla_\mu\nabla_\nu\phi + \frac{1}{2}g_{\mu\nu}\, g^{\rho\sigma}\nabla_\rho\nabla_\sigma\phi\cdot(\ldots) + \textit{connection terms.} \tag{37}$$

Remark 4.3 (Operator-level variation). At the operator level (acting on an arbitrary tensor), the variation $\delta\Box/\delta g^{\mu\nu}$ is itself a differential operator. This is what makes the problem fundamentally different from varying a local action: $\delta F(\Box)/\delta g^{\mu\nu}$ involves an *integral* over the spectrum of $\Box$.

5 The α-Insertion Formula (Barvinsky–Vilkovisky)

This section documents the key technique for varying nonlocal actions.

5.1 The fundamental identity

Theorem 5.1 (α-insertion formula). *[Barvinsky–Vilkovisky, Nucl. Phys. B* ***282*** *(1987) 163; B* ***333*** *(1990) 471]*

Let $F(\Box)$ be an analytic function of the covariant d'Alembertian $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$, and let A, B be scalar fields (or more generally, sections of some vector bundle). Then the

variation of $A\,F(\Box)\,B$ with respect to $g^{\mu\nu}$ is:

$$\boxed{\delta\Big[A\,F(\Box)\,B\Big] = (\delta A)\,F(\Box)\,B + A\,F(\Box)\,(\delta B) + \int_0^1 \mathrm{d}\alpha\,\Big[e^{\alpha\Box}A\Big]\,F'_\alpha(\Box)\,(\delta\Box)\,\Big[e^{(1-\alpha)\Box}B\Big],} \tag{38}$$

where the notation F'_α denotes the "α-deformed derivative" defined via the integral representation:

$$F'_\alpha(\Box) = \left.\frac{F(\Box_1) - F(\Box_2)}{\Box_1 - \Box_2}\right|_{\substack{\Box_1=\alpha\Box\\ \Box_2=(1-\alpha)\Box}}. \tag{39}$$

Remark 5.2 (Simplification for entire functions). If $F(z) = \sum_{n=0}^{\infty} c_n z^n$ is an entire function (as is the case for the SCT form factors, by NT-2), then the α-insertion formula can be written as a sum over the Taylor coefficients:

$$\int_0^1 \mathrm{d}\alpha\,\Big[e^{\alpha\Box}A\Big]\,(\delta\Box)\,\Big[e^{(1-\alpha)\Box}B\Big]\cdot F'_\alpha(\Box) = \sum_{n=1}^{\infty} c_n \sum_{k=0}^{n-1} (\Box^k A)\,(\delta\Box)\,(\Box^{n-1-k}B). \tag{40}$$

This is the "Leibniz rule for $F(\Box)$," generalizing the standard Leibniz rule for $\Box^n$.

Remark 5.3 (Physical meaning). The parameter $\alpha \in [0,1]$ interpolates between "A evaluated at the full nonlocal scale" and "B evaluated at the full nonlocal scale." The α-insertion accounts for the fact that on a curved background, the "nonlocal smearing" generated by $F(\Box)$ interacts with the geometry at all intermediate scales.

[GAP] The precise form of the α-insertion formula for *tensor-valued* form factors (where $\Box$ acts on rank-2 tensors rather than scalars) requires additional curvature correction terms. On a general curved background, the commutator $[\nabla_\mu, \nabla_\nu]V_\rho = R_{\mu\nu\rho}{}^\sigma V_\sigma$ generates Riemann tensor insertions. These must be tracked in the NT-4b derivation.

5.2 Variation of $R\,F(\Box)\,R$

Proposition 5.4 (Variation of the R^2 nonlocal term). *[Deser–Woodard 2007; Nojiri–Odintsov 2008; Calcagni–Modesto 2211.05606]*

The variation of $S_{R^2} = \int \sqrt{g}\,R\,F(\Box)\,R$ with respect to $g^{\mu\nu}$ gives:

$$\begin{aligned}\frac{1}{\sqrt{g}}\frac{\delta S_{R^2}}{\delta g^{\mu\nu}} &= R_{\mu\nu}\,F(\Box)\,R + R\,F(\Box)\,R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}\,R\,F(\Box)\,R\\ &\quad + 2\nabla_\mu\nabla_\nu[F(\Box)\,R] - 2g_{\mu\nu}\Box[F(\Box)\,R]\\ &\quad + g_{\mu\nu}\nabla_\rho[R\,\nabla^\rho F(\Box)\,R] - \nabla_\mu[R\,\nabla_\nu F(\Box)\,R] - \nabla_\nu[R\,\nabla_\mu F(\Box)\,R]\\ &\quad + \Theta^{(R)}_{\mu\nu}[R, F(\Box), R].\end{aligned} \tag{41}$$

The first line contains the "naive" variation of A and B plus the metric determinant. The second line contains the $\delta R/\delta g^{\mu\nu}$ contributions (generating $\nabla\nabla$ and $\Box$ terms). The third and fourth lines contain the α-insertion terms $\Theta^{(R)}_{\mu\nu}$ from $\delta\Box/\delta g^{\mu\nu}$:

$$\Theta^{(R)}_{\mu\nu} = -\sum_{n=1}^{\infty} c_n \sum_{k=0}^{n-1}\Big[\nabla_\mu(\Box^k R)\,\nabla_\nu(\Box^{n-1-k}R) - \frac{1}{2}g_{\mu\nu}\,\nabla_\rho(\Box^k R)\,\nabla^\rho(\Box^{n-1-k}R)\Big], \tag{42}$$

where $F(z) = \sum c_n z^n$. In closed form (using the α-representation):

$$\Theta^{(R)}_{\mu\nu} = -\int_0^1 \mathrm{d}\alpha\,\Big[\nabla_\mu\Phi_\alpha\,\nabla_\nu\Psi_{1-\alpha} - \frac{1}{2}g_{\mu\nu}\,\nabla_\rho\Phi_\alpha\,\nabla^\rho\Psi_{1-\alpha}\Big], \tag{43}$$

where $\Phi_\alpha = e^{\alpha\Box}R$ and $\Psi_{1-\alpha} = F'_\alpha(\Box)\,e^{(1-\alpha)\Box}R$.

Remark 5.5 (Local limit check). When $F(\Box) \to \beta$ (constant), all the Θ terms vanish (since $F' = 0$ for constant F), and we recover:

$$\frac{1}{\sqrt{g}} \frac{\delta}{\delta g^{\mu\nu}} \int \sqrt{g}\, \beta R^2 = 2\beta\, H_{\mu\nu}, \tag{44}$$

which matches (19). This is a mandatory consistency check.

5.3 Variation of $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} F(\Box)\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ (nonlocal Bach)

The variation of the Weyl-squared term is more involved because the Weyl tensor has four indices.

Proposition 5.6 (Variation of the nonlocal Weyl term). *[Stelle 1978 (local limit); Calcagni–Modesto 2211.05606 (nonlocal case); Giacchini–de Paula Netto 2021]*

Schematically, the variation of $S_{C^2} = \int \sqrt{g}\, C_{\mu\nu\rho\sigma}\, F(\Box)\, C^{\mu\nu\rho\sigma}$ yields:

$$\begin{aligned} \frac{1}{\sqrt{g}} \frac{\delta S_{C^2}}{\delta g^{\mu\nu}} &= 2\, B^{(F)}_{\mu\nu} - \frac{1}{2} g_{\mu\nu}\, C_{\alpha\beta\gamma\delta}\, F(\Box)\, C^{\alpha\beta\gamma\delta} \\ &\quad + \Theta^{(C)}_{\mu\nu}[C_{\rho\sigma\alpha\beta}, F(\Box), C^{\rho\sigma\alpha\beta}], \end{aligned} \tag{45}$$

*where $B^{(F)}_{\mu\nu}$ is the **nonlocal Bach tensor**:*

$$B^{(F)}_{\mu\nu} = \nabla^\rho \nabla^\sigma \Big[F(\Box)\, C_{\mu\rho\nu\sigma}\Big] + \frac{1}{2} R^{\rho\sigma}\, F(\Box)\, C_{\mu\rho\nu\sigma} + (\textit{mixed terms}), \tag{46}$$

generalizing the local Bach tensor (18) by replacing the curvature C with $F(\Box)\, C$.

The $\Theta^{(C)}_{\mu\nu}$ terms are the α-insertion contributions from the variation of $\Box$ inside $F(\Box)$ when it acts on the four-index Weyl tensor. These are structurally analogous to (42) but involve the full four-index tensor structure.

[GAP] The explicit closed-form expression for $\Theta^{(C)}_{\mu\nu}$ is not available in a single reference in the $\{C^2, R^2\}$ basis. Most references work in the $\{R^2_{\mu\nu}, R^2\}$ or $\{G_{\mu\nu}, R^{\mu\nu}\}$ basis (see Calcagni–Modesto [11]), and the translation to the Weyl basis requires using (7) plus the Gauss–Bonnet identity. This translation must be performed carefully in the NT-4b derivation.

6 Nonlocal Gravity in the $G_{\mu\nu}\gamma(\Box)R^{\mu\nu}$ Basis

An alternative approach, used extensively in the IDG literature, writes the action in terms of the Einstein tensor.

6.1 The simplified action

Convention 6.1 (Calcagni–Modesto simplified action). [Calcagni–Modesto 2211.05606, Eq. (simplifiedNLGActionF)] With the choice $\gamma_2 = -2\gamma_0 \equiv \gamma(\Box)$ and $\gamma_4 = 0$:

$$S_{\rm NLG} = \frac{1}{2\kappa^2} \int \mathrm{d}^4x\, \sqrt{|g|}\, \Big[R + G_{\mu\nu}\, \gamma(\Box)\, R^{\mu\nu}\Big]. \tag{47}$$

Theorem 6.2 (Equations of motion in the $G\gamma R$ basis)**.** *[Calcagni–Modesto 2211.05606, Eq. (EOMsNLG); Calcagni–Modesto–Leszczyszyn 2018] The field equations derived from (47) are:*

$$\begin{aligned}\kappa^2 T_{\mu\nu} = e^{-r_*\Box} G_{\mu\nu} &- \frac{1}{2} g_{\mu\nu}\, G_{\rho\sigma}\, \gamma(\Box)\, R^{\rho\sigma} + 2G^{\rho}{}_{(\mu}\, \gamma(\Box)\, G_{\nu)\rho} \\ &+ g_{\mu\nu} \nabla^\rho \nabla^\sigma \gamma(\Box)\, G_{\rho\sigma} - 2\nabla^\rho \nabla_{(\mu} \gamma(\Box)\, G_{\nu)\rho} \\ &+ \frac{1}{2}\Big(G_{\mu\nu}\, \gamma(\Box)\, R + R\, \gamma(\Box)\, G_{\mu\nu}\Big) \\ &+ \Theta_{\mu\nu}(R_{\rho\sigma}, G^{\rho\sigma}),\end{aligned} \tag{48}$$

where $\Theta_{\mu\nu}$ contains the α-insertion contributions. For the Wataghin/minimal form factor $\gamma(\Box) = (e^{-r_*\Box} - 1)/\Box$:

$$\Theta_{\mu\nu}(R_{\rho\sigma}, G^{\rho\sigma}) = -\int_0^{r_*} \mathrm{d}q\, \tilde{\Theta}_{\mu\nu}\left[e^{-q\Box} R_{\rho\sigma},\ \frac{e^{-(r_*-q)\Box} - 1}{\Box}\, G^{\rho\sigma}\right], \tag{49}$$

with:

$$\tilde{\Theta}^{\mathrm{sym}}_{\mu\nu}(A_{\rho\sigma}, B^{\rho\sigma}) = -\nabla_\mu A_{\rho\sigma}\, \nabla_\nu B^{\rho\sigma} + \frac{1}{4} g_{\mu\nu}\, \nabla_\tau (A_{\rho\sigma} \nabla^\tau B^{\rho\sigma} + B^{\rho\sigma} \nabla^\tau A_{\rho\sigma}), \tag{50}$$

$$\begin{aligned}\tilde{\Theta}^{\mathrm{antisym}}_{\mu\nu}(A_{\rho\sigma}, B^{\rho\sigma}) = \frac{1}{4} g_{\mu\nu}\, &\nabla_\tau (A_{\rho\sigma} \nabla^\tau B^{\rho\sigma} - B^{\rho\sigma} \nabla^\tau A_{\rho\sigma}) \\ &+ \nabla_\rho (A_{\mu\sigma} \nabla^\rho B^\sigma{}_\nu - B^\sigma{}_\nu \nabla^\rho A_{\mu\sigma}) \\ &+ \nabla_\sigma (B_{\mu\rho} \nabla_\nu A^{\rho\sigma} - A^{\rho\sigma} \nabla_\nu B_{\mu\rho}) \\ &+ \nabla_\rho (A_{\mu\sigma} \nabla_\nu B^{\sigma\rho} - B^{\sigma\rho} \nabla_\nu A_{\mu\sigma}).\end{aligned} \tag{51}$$

Here indices μ and ν are implicitly symmetrized.

Remark 6.3 (Recovery of Einstein's equations). Setting $\gamma(\Box) = 0$ (i.e., $r_* = 0$), all nonlocal terms vanish and (48) reduces to $G_{\mu\nu} = \kappa^2 T_{\mu\nu}$.

Remark 6.4 (Ricci-flat solutions). [Calcagni–Modesto 2211.05606, § SecRicciFlat] If $R_{\mu\nu} = 0$ (hence $G_{\mu\nu} = 0$), then all terms in (48) vanish and the vacuum Einstein solutions (Schwarzschild, Kerr) remain exact solutions of the nonlocal theory when $\gamma_4 = 0$. This is an important structural property.

6.2 Mapping to the SCT $\{C^2, R^2\}$ basis

Proposition 6.5 (Basis translation)**.** *[needs check] The SCT action (4) in the $\{C^2, R^2\}$ basis maps to the general action (8) via:*

$$\gamma_4 = \frac{1}{16\pi^2} F_1\left(\frac{\Box}{\Lambda^2}\right), \tag{52}$$

$$\gamma_2 = -\frac{2}{16\pi^2} F_1\left(\frac{\Box}{\Lambda^2}\right), \tag{53}$$

$$\gamma_0 = \frac{1}{16\pi^2}\left[F_2\left(\frac{\Box}{\Lambda^2}, \xi\right) + \frac{1}{3} F_1\left(\frac{\Box}{\Lambda^2}\right)\right]. \tag{54}$$

Note that the Calcagni–Modesto simplified form (47) uses $\gamma_4 = 0$, which does not *match the SCT action (where $\gamma_4 \neq 0$ corresponds to the Weyl term). Therefore, the equations (48) cannot be used directly for the SCT case. Instead, one must either:*

(a) Use the general action (8) with all three form factors, or
(b) Work directly in the $\{C^2, R^2\}$ basis and vary each term separately.
We recommend approach (b) for NT-4b, as it maintains the clean spin-2/spin-0 separation throughout.

7 Key Identities for the Nonlinear Variation

7.1 Commutator of $\Box$ with curvature tensors

Proposition 7.1 (Commutation relations). *[Avramidi, "Heat Kernel and Quantum Gravity" (2000), Ch. 3; Parker–Toms Ch. 6] On curved backgrounds, the covariant d'Alembertian does not commute with curvature tensors. The key relations are:*

$$[\nabla_\mu, \nabla_\nu]\phi = 0 \quad \textit{(scalar)}, \tag{55}$$
$$[\nabla_\mu, \nabla_\nu]V_\rho = R_{\mu\nu\rho}{}^{\sigma} V_\sigma, \tag{56}$$
$$[\nabla_\mu, \nabla_\nu]T_{\rho\sigma} = R_{\mu\nu\rho}{}^{\alpha} T_{\alpha\sigma} + R_{\mu\nu\sigma}{}^{\alpha} T_{\rho\alpha}, \tag{57}$$
$$[\Box, \nabla_\mu]\phi = R_{\mu\nu}\nabla^\nu \phi, \tag{58}$$
$$[\Box, \nabla_\mu]V_\nu = R_{\mu\rho}\nabla^\rho V_\nu + R_{\mu\nu}{}^{\rho\sigma}\nabla_\rho V_\sigma + (\nabla^\rho R_{\mu\rho\nu}{}^{\sigma})V_\sigma. \tag{59}$$

Remark 7.2 (Implications for NT-4b). The commutator $[\Box, \nabla]$ generates Ricci and Riemann insertions every time one moves $\Box$ past a covariant derivative. In the α-insertion formula, this means the $\Theta_{\mu\nu}$ terms receive curvature corrections at every order. For the SCT form factors (which are entire functions), these corrections are automatically encoded in the integral representation (43).

7.2 The Lichnerowicz operator

Definition 7.3 (Lichnerowicz operator). [Calcagni–Modesto 2211.05606; Lichnerowicz 1961] Acting on a symmetric rank-2 tensor $\psi_{\mu\nu}$:

$$\Delta_L \psi_{\mu\nu} = 2R^\rho{}_{\mu\nu}{}^{\tau}\psi_{\tau\rho} + R_{\mu\rho}\psi^\rho{}_\nu + R_{\sigma\nu}\psi^\sigma{}_\mu - \Box\psi_{\mu\nu}. \tag{60}$$

On a scalar: $\Delta_L \psi = -\Box\psi$.

Remark 7.4 (Role in stability). [Calcagni–Modesto 2211.05606, § stabi] The Lichnerowicz operator encodes the curvature corrections to $\Box$ when acting on tensors. In the nonlocal theory, replacing $-\Box \to \Delta_L$ in the form factor gives a theory $\mathcal{L} = R + G_{\mu\nu}\gamma(\Delta_L)R^{\mu\nu}$ that shares the same linearized spectrum on Minkowski and inherits stability of Ricci-flat solutions to all perturbative orders.

7.3 Integration by parts with $F(\Box)$

Proposition 7.5 (Self-adjointness of $\Box$). *[exact from lit.] On a closed manifold (or with vanishing boundary terms):*

$$\int \sqrt{g}\, A\, \Box B\, \mathrm{d}^4x = \int \sqrt{g}\, (\Box A)\, B\, \mathrm{d}^4x = -\int \sqrt{g}\, \nabla_\mu A\, \nabla^\mu B\, \mathrm{d}^4x. \tag{61}$$

More generally, for an entire function $F(\Box)$:

$$\int \sqrt{g}\, A\, F(\Box)\, B\, \mathrm{d}^4x = \int \sqrt{g}\, \big[F(\Box)\, A\big]\, B\, \mathrm{d}^4x. \tag{62}$$

This self-adjointness is crucial for simplifying the field equations: terms like $R_{\mu\nu}\, F(\Box)\, R + R\, F(\Box)\, R_{\mu\nu}$ can be symmetrized.

8 Consistency Checks for the Full Nonlinear Equations

8.1 Bianchi identity

Proposition 8.1 (Diffeomorphism invariance). *[exact from lit.] The full nonlinear equations must satisfy:*

$$\nabla^\mu \left(\frac{1}{\sqrt{g}} \frac{\delta S}{\delta g^{\mu\nu}} \right) = 0, \tag{63}$$

identically (off-shell), as a consequence of the diffeomorphism invariance of S. This is the fully nonlinear contracted Bianchi identity.

In the local (Stelle) limit, this reduces to $\nabla^\mu (G_{\mu\nu} + 2\alpha B_{\mu\nu} + 2\beta H_{\mu\nu}) = 0$, which holds because $\nabla^\mu G_{\mu\nu} = 0$, $\nabla^\mu B_{\mu\nu} = 0$, and $\nabla^\mu H_{\mu\nu} = 0$ individually. For the nonlocal case, the α-insertion terms $\Theta_{\mu\nu}$ must also be covariantly conserved:

$$\nabla^\mu \Theta_{\mu\nu} = 0. \tag{64}$$

8.2 Trace equation

Proposition 8.2 (Conformal anomaly from the trace). *[Deser–Duff–Isham 1976; Duff 1994] The trace $g^{\mu\nu}(\delta S/\delta g^{\mu\nu})$ of the field equations in the local limit gives:*

$$g^{\mu\nu}(G_{\mu\nu} + 2\alpha B_{\mu\nu} + \beta H_{\mu\nu}) = -R + 0 + \beta(-6\Box R) = -R - 6\beta\Box R. \tag{65}$$

The Bach tensor contribution vanishes ($B_{\mu\nu}$ is traceless). In the nonlocal case, the trace should reproduce the conformal anomaly structure when evaluated on conformally flat backgrounds.

8.3 Linearized limit

Proposition 8.3 (Recovery of NT-4a). *[our convention] When the full nonlinear equations are linearized around flat space ($g_{\mu\nu} = \delta_{\mu\nu} + h_{\mu\nu}$, $|h| \ll 1$), all nonlinear curvature terms (including the $\Theta_{\mu\nu}$ contributions) must vanish or reduce to their linearized forms, reproducing exactly:*

$$\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z), \tag{66}$$

$$\Pi_s(z,\xi) = 1 + 6\left(\xi - \frac{1}{6}\right)^2 z\, \hat{F}_2(z,\xi). \tag{67}$$

This is the most important consistency check: NT-4b must contain NT-4a as a special case.

8.4 De Sitter solutions

Proposition 8.4 (Maximally symmetric solutions). *[Calcagni–Modesto 2211.05606, § sectionRF; Biswas–Mazumdar–Siegel 2006] On a maximally symmetric background (de Sitter or anti-de Sitter):*

$$R_{\mu\nu\rho\sigma} = \frac{R_0}{12}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}), \quad R_{\mu\nu} = \frac{R_0}{4}g_{\mu\nu}, \quad R = R_0 = const. \tag{68}$$

The Weyl tensor vanishes: $C_{\mu\nu\rho\sigma} = 0$. Therefore the $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}F(\Box)C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ term contributes nothing, and the field equations reduce to:

$$G_{\mu\nu} + \beta\, H_{\mu\nu}\Big|_{dS} = \kappa^2 T_{\mu\nu}, \tag{69}$$

where $H_{\mu\nu}|_{dS}$ involves only constant-curvature contributions. Since $\Box R_0 = 0$ and $\nabla_\mu R_0 = 0$, the $H_{\mu\nu}$ tensor simplifies to:

$$H_{\mu\nu}\Big|_{dS} = -\frac{1}{2}g_{\mu\nu}R_0^2 + 2R_0 \cdot \frac{R_0}{4}g_{\mu\nu} = 0. \tag{70}$$

Therefore, the de Sitter condition becomes simply $G_{\mu\nu} = \kappa^2 T_{\mu\nu}$ (or $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$ in vacuum). De Sitter spacetime with the correct cosmological constant is an exact solution of the full nonlocal theory.

9 Auxiliary Field Formulation

An alternative approach to deriving the field equations uses auxiliary fields to localize the nonlocal action.

Proposition 9.1 (Auxiliary field method). *[Calcagni–Modesto 2211.05606, Eq. (AuxAction)] Introduce a symmetric tensor $\phi_{\mu\nu}$ and write:*

$$\tilde{S}[g,\phi] = \frac{1}{2\kappa^2}\int \mathrm{d}^4x\,\sqrt{|g|}\,\left[R + (2R_{\mu\nu} - \phi_{\mu\nu} + \frac{1}{2}g_{\mu\nu}\phi)\,\gamma(\Box)\,\phi^{\mu\nu}\right], \tag{71}$$

where $\phi = g^{\mu\nu}\phi_{\mu\nu}$. The $\phi_{\mu\nu}$ equation of motion yields $\phi_{\mu\nu} = G_{\mu\nu}$ on-shell, recovering the original nonlocal action.

The advantage is that the $g_{\mu\nu}$ variation of $\tilde{S}$ is algebraically simpler *because the nonlocal operator $\gamma(\Box)$ now acts on the auxiliary field rather than directly on curvature tensors.*

Remark 9.2 (Degrees of freedom). [Calcagni–Modesto 2211.05606] The auxiliary field $\phi_{\mu\nu}$ is a symmetric rank-2 tensor (10 components) constrained by the Bianchi identity (4 constraints) to have 6 degrees of freedom. Combined with the 2 degrees of freedom of the graviton, the total is $2 + 6 = 8$ degrees of freedom, matching Stelle's gravity counting.

In the nonlocal theory, the key difference is that the additional 6 degrees of freedom are *not ghosts*: the propagator $\propto \gamma\, e^{-H(\Box)}$ has no additional poles when γ is an entire function of appropriate type.

10 Notation Table

Symbol	Definition	Source
$g_{\mu\nu}$	spacetime metric, signature $(-,+,+,+)$	Standard
$R^{\rho}{}_{\sigma\mu\nu}$	Riemann tensor $(\partial_\mu \Gamma^{\rho}_{\nu\sigma} - \ldots)$	Standard
$R_{\mu\nu}$	Ricci tensor $(R^{\rho}{}_{\mu\rho\nu})$	Standard
R	Ricci scalar $(g^{\mu\nu} R_{\mu\nu})$	Standard
$G_{\mu\nu}$	Einstein tensor $(R_{\mu\nu} - \frac{1}{2} g R)$	Standard
$C_{\mu\nu\rho\sigma}$	Weyl tensor	Standard
E_4	Gauss–Bonnet density	Eq. (6)
$\Box$	$g^{\mu\nu}\nabla_\mu \nabla_\nu$ (d'Alembertian)	Standard
Λ	spectral action cutoff scale	SCT P3
$\Lambda_{\rm cc}$	cosmological constant	$S_{\rm EH}$
κ^2	$8\pi G$ (gravitational coupling)	Standard
$F_1(z)$	Weyl form factor $(z = \Box/\Lambda^2)$	NT-1b
$F_2(z,\xi)$	R^2 form factor	NT-1b
$\hat{F}_i(z)$	normalized form factor $(\hat{F}_i(0) = 1)$	NT-4a
α_C	$16\pi^2 F_1(0) = 13/120$	Phase 3
$\alpha_R(\xi)$	$16\pi^2 F_2(0) = 2(\xi - 1/6)^2$	Phase 3
c_1, c_2	Stelle basis coefficients	Phase 3
$\varphi(z)$	master function $e^{-z/4}\sqrt{\pi/z}\ \mathrm{erfi}(\sqrt{z}/2)$	NT-1
ξ	Higgs non-minimal coupling	SM
$B_{\mu\nu}$	Bach tensor (local)	Eq. (18)
$B^{(F)}_{\mu\nu}$	nonlocal Bach tensor	Eq. (46)
$H_{\mu\nu}$	scalar curvature tensor (local)	Eq. (19)
$\Theta_{\mu\nu}$	α-insertion contribution	BV formalism
$\gamma_0, \gamma_2, \gamma_4$	form factors in Riemann basis	CMN
Δ_L	Lichnerowicz operator	Eq. (60)
$\Pi_{\rm TT}(z)$	spin-2 propagator denominator	NT-4a
$\Pi_s(z,\xi)$	spin-0 propagator denominator	NT-4a
m_2	$\Lambda\sqrt{60/13}$ (spin-2 mass)	NT-4a
m_0	$\Lambda/\sqrt{6(\xi - 1/6)^2}$ (scalar mass)	NT-4a
N_s, N_f, N_v	SM multiplicities $(4, 45, 12)$	CPR
N_D	Dirac fermion count $(N_f/2 = 22.5)$	Phase 3

11 Gap Analysis

G1. Explicit $\Theta^{(C)}_{\mu\nu}$ in the Weyl basis. The α-insertion terms for the Weyl-squared variation $\delta(C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}F(\Box)C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma})/\delta g^{\mu\nu}$ are not available in closed form in a single published reference in the $\{C^2, R^2\}$ basis. The Calcagni–Modesto result is in the $\{G_{\mu\nu}, R^{\mu\nu}\}$ basis. Translation requires careful application of the Weyl decomposition and Gauss–Bonnet identity.

G2. Tensor-valued α-insertion. The α-insertion formula (Theorem 5.1) is stated for scalar-valued quantities A, B. When $\Box$ acts on rank-4 tensors (as in $F(\Box)\,C_{\mu\nu\rho\sigma}$), the commutator $[\Box, \nabla]$ generates Riemann tensor insertions at every order. The systematic treatment requires the full Barvinsky–Vilkovisky covariant perturbation theory to second order in curvature.

G3. Non-commutativity of F_1 and F_2 form factors. In the SCT action, F_1 and F_2 are different functions of $\Box/\Lambda^2$. When varying $g^{\mu\nu}$, the variations of $F_1(\Box)$ and $F_2(\Box)$ generate different Θ-type terms. These must be computed separately and summed.

G4. Boundary terms. The self-adjointness identity (62) assumes vanishing boundary contributions. For the FLRW reduction (NT-4c), boundary terms at the initial singularity or at spatial infinity may need careful treatment.

G5. Sign convention propagation. The Stelle equations (17) use the convention $G_{\mu\nu} + 2\alpha B_{\mu\nu} + \beta H_{\mu\nu} = \kappa^2 T_{\mu\nu}$. The mapping $\alpha \to F_1(0)/(16\pi^2)$, $\beta \to F_2(0)/(16\pi^2)$ must include all factors of 2 from the Weyl decomposition. This is the most common source of sign errors in the literature.

G6. The Gauss–Bonnet term with nonlocal form factor. If one works in the full $\{R^2_{\mu\nu\rho\sigma}, R_{\mu\nu}R^{\mu\nu}, R^2\}$ basis, the Gauss–Bonnet topological identity $\delta(\sqrt{g}\,E_4)/\delta g^{\mu\nu} = 0$ holds only when the form factor is unity (local case). For $\int \sqrt{g}\,R_{\mu\nu\rho\sigma}\,\gamma_4(\Box)\,R^{\mu\nu\rho\sigma}$ with $\gamma_4 \neq$ const, the Gauss–Bonnet cancellation does *not* hold because $[\gamma_4(\Box), \delta/\delta g^{\mu\nu}] \neq 0$. This means the basis reduction must be done *before* applying the form factor, not after.

G7. Connection to the spectral action. The form factors F_1 and F_2 are derived from the heat kernel expansion of $\mathrm{Tr}(f(D^2/\Lambda^2))$. The nonlocal effective action is obtained by resumming the heat kernel to second order in curvature (Barvinsky–Vilkovisky CPT-II). The field equations are the Euler–Lagrange equations of this resummed action. The consistency between the heat-kernel resummation and the variational equations must be verified.

12 Strategy for the NT-4b Derivation

Based on the literature survey, we recommend the following strategy for the NT-4b derivation:

Step 1. Work in the $\{C^2, R^2\}$ basis. This maintains the clean spin-2/spin-0 separation throughout and allows direct comparison with the NT-4a linearized results.

Step 2. Vary S_{R^2} and S_{C^2} separately. Use the formulas from §4 for each term. The R^2 variation is cleaner (scalar × scalar) and should be done first as a warm-up.

Step 3. Use the α-insertion formula for Θ terms. For each form factor, compute the α-insertion contribution using the Taylor expansion (40) (which is valid because the SCT form factors are entire by NT-2).

Step 4. Verify the local limit. Set $F_i \to F_i(0)$ and confirm recovery of the Stelle equations (17) with the correct coefficients $\alpha = \alpha_C/(16\pi^2)$, $\beta = \alpha_R/(16\pi^2)$.

Step 5. Verify the Bianchi identity. Check $\nabla^\mu(\delta S/\delta g^{\mu\nu}) = 0$ at the full nonlinear level.

Step 6. Linearize and recover NT-4a. Expand around flat space and verify recovery of $\Pi_{\rm TT}$ and Π_s.

Step 7. Check de Sitter and Ricci-flat solutions. Verify that de Sitter spacetime and Schwarzschild remain solutions.

Step 8. Write the final field equations in canonical form. Present the result as:

$$G_{\mu\nu} + 2\alpha\, B^{(F_1)}_{\mu\nu} + \beta\, H^{(F_2)}_{\mu\nu} + \Theta^{(1)}_{\mu\nu} + \Theta^{(2)}_{\mu\nu} = \kappa^2 T_{\mu\nu}. \tag{72}$$

References

[1] K. S. Stelle, "Renormalization of higher-derivative quantum gravity," Phys. Rev. D **16** (1977) 953.

[2] K. S. Stelle, "Classical gravity with higher derivatives," Gen. Rel. Grav. **9** (1978) 353–371.

[3] A. O. Barvinsky, G. A. Vilkovisky, "The generalized Schwinger-DeWitt technique in gauge theories and quantum gravity," Phys. Rep. **119** (1985) 1–74.

[4] A. O. Barvinsky, G. A. Vilkovisky, "Beyond the Schwinger-DeWitt technique: Converting loops into trees and in-in currents," Nucl. Phys. B **282** (1987) 163–188.

[5] A. O. Barvinsky, G. A. Vilkovisky, "Covariant perturbation theory (II): Second order in the curvature. General algorithms," Nucl. Phys. B **333** (1990) 471–511.

[6] A. O. Barvinsky, G. A. Vilkovisky, "Covariant perturbation theory (III): Spectral representations of the third-order form factors," Nucl. Phys. B **333** (1990) 512–524.

[7] A. Codello, O. Zanusso, "On the non-local heat kernel expansion," J. Math. Phys. **54** (2013) 013513, arXiv:1203.2034 [hep-th].

[8] A. Codello, R. Percacci, C. Rahmede, "Investigating the ultraviolet properties of gravity with a Wilsonian renormalization group equation," Ann. Phys. **324** (2009) 414, arXiv:0805.2909 [hep-th].

[9] T. Biswas, A. Mazumdar, W. Siegel, "Bouncing universes in string-inspired gravity," JCAP **0603** (2006) 009, hep-th/0508194.

[10] L. Modesto, "Super-renormalizable quantum gravity," Phys. Rev. D **86** (2012) 044005, arXiv:1107.2403 [hep-th].

[11] G. Calcagni, L. Modesto, "Classical and quantum nonlocal gravity," arXiv:2211.05606 [hep-th] (2022).

[12] G. Calcagni, L. Modesto, G. Nardelli, "Initial conditions and degrees of freedom of non-local gravity," JHEP **05** (2018) 087, arXiv:1803.00561 [hep-th].

[13] G. Calcagni, L. Modesto, G. Nardelli, "Non-perturbative spectrum of non-local gravity," arXiv:1803.07848 [hep-th].

[14] L. Buoninfante, G. Lambiase, Y. Miyashita, W. Takebe, M. Yamaguchi, "Generalized ghost-free propagators in nonlocal field theories," Phys. Rev. D **101** (2020) 084019, arXiv:2001.07830 [hep-th].

[15] B. L. Giacchini, T. de Paula Netto, "Higher-order regularity in local and nonlocal quantum gravity," Eur. Phys. J. C **81** (2021) 741.

[16] B. L. Giacchini, T. de Paula Netto, "Effective delta sources and regularity in higher-derivative and ghost-free gravity," JCAP **07** (2019) 013, arXiv:1806.05664 [gr-qc].

[17] E. T. Tomboulis, "Super-renormalizable gauge and gravitational theories," hep-th/9702080 (1997).

[18] I. G. Avramidi, "Heat Kernel and Quantum Gravity," Springer, Lecture Notes in Physics **M64** (2000).

[19] L. Parker, D. Toms, "Quantum Field Theory in Curved Spacetime: Quantized Fields and Gravity," Cambridge University Press (2009).

[20] F. Briscese, G. Calcagni, L. Modesto, "Nonlinear stability in nonlocal gravity," Phys. Rev. D **99** (2019) 084041, arXiv:1901.03267 [gr-qc].

[21] S. Deser, R. P. Woodard, "Nonlocal cosmology," Phys. Rev. Lett. **99** (2007) 111301.

[22] A. H. Chamseddine, A. Connes, "The spectral action principle," Commun. Math. Phys. **186** (1997) 731, hep-th/9606001.

[23] D. Alfyorov, "NT-4a: Literature extraction & convention fixing for linearized nonlocal field equations," SCT Theory internal document (2026).

[24] T. Biswas, E. Gerber, A. Mazumdar, "Towards singularity- and ghost-free theories of gravity," Phys. Rev. Lett. **108** (2012) 031101, arXiv:1112.5430 [gr-qc].